

VICTORIA MCCRAY

Biomedical & Community Health Data Scientist | MSc in Bioinformatics

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RESEARCH INTERESTS

Biomedical & Community Health Data Scientist passionate about research within neuroinformatics, ethics, open science and data science outreach. Over five years developing statistical and machine learning solutions across biotechnology, medical and wearable devices, public health, and healthcare. Experienced in analyzing neuroimaging and functional network connectivity, clinical and epidemiological surveillance data, behavioral and cognitive assessments, genomic sequences, product usage patterns, and pharmacological outcomes. Expertise applying Python, R, and cloud platforms to extract insights from complex, high-dimensional biological data. Recognized for technical documentation, cross-sector collaboration, and building open science capacity in resource-limited settings.

EDUCATION

Master of Science, Bioinformatics | *Northeastern University* | May 2026

Focus: Neuroimaging & Biomedical Data Science, Bioinformatics Pipelines, ML Evaluation

Bachelor of Science, Psychology (Honors) | *Northeastern University* | May 2022

Minor: Data Science | Honors Thesis: Effects of Cannabis for Persons with Schizophrenia: Implications for Reward Processing (<https://repository.library.northeastern.edu/files/neu:4f171j65d>)

PUBLICATIONS & PRESENTATIONS

[Journal Article]

McCray, V.P., Dukes, A., Pittman, Nikea. *Community-Driven Programming to Strengthen STEM Conference Experiences: A First Look at BlackInNeuro and BlackInMicro Event Outcomes*. *Oxford Open Neuro*. (in press, 2026).

[Government Report]

Contributor. *Centers for Disease Control and Prevention. Drug Overdose Surveillance and Epidemiology Discharge Surveillance (DOSE-DIS) System: Nonfatal Overdose Emergency Department and Inpatient Hospitalization Discharge Data*. US Department of Health and Human Services, CDC; 2024. Access at: <https://www.cdc.gov/overdose-prevention/data-research/facts-stats/dose-dashboard-nonfatal-discharge-data.html>

[Conference Presentations]

Makwakwa, D., McCray, V.P. (2026) *Community-Based Digital Screener for Early Dementia Detection in Malawi*. *Alzheimer's Association International Conference*. London, UK.

Dukes, A., Murray, D., Kelly, C., Greenwood, P., McCray, V., et al. (2023). *How I Survived Grad School: Perspectives from Black In Neuro*. *Society for Neuroscience 2023*. Washington D.C.

McCray, V. (2022). *Effects of Cannabis for Persons with Schizophrenia: Implications for Reward Processing*. *RISE Expo Poster Session*. Boston, MA.

GRANTS, SCHOLARSHIPS & FELLOWSHIPS

Graduate Student Scholarship

May 2026

College of Science, Northeastern University

Sep 2023 -

Awarded to support tuition for Master's coursework in Bioinformatics.

Srinivasan Family Award

Dec 2023

Center for Emerging Markets, Northeastern University

Grant awarded to develop open-source research tools for processing and analyzing neuroimaging data in emerging markets.

Paul and Grace Ward Martinez ('48) Research Fellowship

Sep 2021

Northeastern University

Supported undergraduate research in the Learning and Brain Development Lab.

HONORS & AWARDS

Semi-Finalist, College of Science Dean's Excellence Award for Experiential Learning

Mar 2026

Northeastern University

Honors in Psychology

May 2022

Northeastern University

Completed undergraduate honors thesis in psychology and cognitive neuroscience under the supervision of Juliet Davidow, PhD.

Avrom Aaron Leve Memorial Award

May 2022

Northeastern University

Selected in recognition of academic achievements, personal character, and commitment to service.

Faculty-Selected Senior Award

May 2022

Dept. of Psychology, Northeastern University

Selected by faculty in recognition of academic achievement among graduating seniors.

RESEARCH EXPERIENCE

Founder & Lead Facilitator | *Open Science Neuro Hackathon* | Feb 2025 | Lilongwe, Malawi

Collaborators: Reality Labs at Meta, African Brain Data Network, Black In Neuro, TReND in Africa, Lilongwe University of Agriculture & Natural Resources, Learning & Brain Development Lab at NEU

- ▶ Founded international neuroimaging and data science program for 30+ graduate students and faculty, teaching Python and computational neuroscience in resource-limited settings
- ▶ Facilitated curriculum on biomedical imaging AI including medical image classification, tumor detection from sMRI scans, and CNNs for diagnostic imaging
- ▶ Contributed documentation improvements to nilearn, an open-source Python library for neuroimaging analysis

Research Fellow | *Learning & Brain Development Lab, Northeastern University* | Sep 2021 - May 2022 | Boston, MA

- ▶ Conducted behavioral and network analysis of brain connectivity in clinical datasets using Python, R, and CONN toolbox for functional connectivity analysis
- ▶ Performed hypothesis testing, ANOVA, and regression to investigate cognitive and neurological development in psychiatric populations

- ▶ Developed computational pipelines for processing and analyzing fMRI data from clinical datasets

Research Assistant | *Interdisciplinary Affective Science Lab, Northeastern University* | April - Dec 2019 | Boston, MA

- ▶ Scored physiological data for 10-12 participants weekly (heart rate variability, skin conductance, electrical impedance)
- ▶ Assisted in presenting research, developing abstracts, and contributing to poster presentations
- ▶ Thematically coded transcripts of qualitative data

Research Assistant | *Phonology & Reading (Language & Mind) Lab, Northeastern University* | Sep 2018 - Apr 2019

- ▶ Organized, recruited, and conducted 10-15 experiments per week for deaf and hearing individuals
- ▶ Attended lab meetings and contributed to experimental design
- ▶ Processed data entry, data coding, and descriptive statistics

PROFESSIONAL EXPERIENCE

Public Health AI & Data Scientist II | *Guidehouse (Supporting the CDC)* | Feb 2024 – May 2026 | Atlanta, GA

- ▶ Architected enterprise data infrastructure (PySpark/Databricks) processing 11M+ records for national overdose surveillance across 54 jurisdictions
- ▶ Designed and deployed transformer-based recommendation system (OpenAI API) for automated document analysis for reducing manual research time
- ▶ Designed and formatted 15+ data visualizations for two peer-reviewed manuscripts on national Parkinson's Disease and Multiple Sclerosis prevalence using clinical surveillance data
- ▶ Led responsible AI adoption strategy via developed curriculum, use cases, and governance frameworks for C-suite leadership
- ▶ Established Python/R SME technical standards with Git workflows and code review for production QA

UX Researcher / Data Analyst | *Autodesk* | Feb 2022 – Apr 2023 | Boston, MA

- ▶ Designed product usage analytics framework (Looker, Qualtrics XM) surfacing user segments and feature adoption patterns for product leadership
- ▶ Led NLP and statistical modeling (regression, clustering) on 1M+ survey responses to directly influence product roadmap

UX Researcher | *Insulet Corporation* | Jan 2021 – Jun 2021 | Billerica, MA

- ▶ Designed and analyzed A/B user tests for the wearable medical device and automated insulin controller app
- ▶ Managed full usability study lifecycle including statistical analysis and Qualtrics platform reporting

SERVICE & LEADERSHIP

Ethics Reviewer | *NeurIPS* | 2026 - Present

- ▶ Provide feedback to the program committee regarding risks and harms of the work in line with the NeurIPS Code of Ethics

Director of Finance | *Black In Neuro* | 2023 - 2026

Previously: Reporting and Finance Lead (2021-2023)

- ▶ Directed financial planning and impact reporting for the international 501(c)(3) nonprofit supporting Black neuroscientists across 40+ countries with \$150K annual funding portfolio

- ▶ Increased funding diversification 35% through donor tracking, in-kind support, and strategic sponsor outreach

Legacy Mentor | *John D. O’Bryant African American Institute* | Jan 2024 - April 2024

- ▶ Provided informal mentorship to undergraduate students for scholarships, networking, and academic resources

Development Intern | *New American Pathways* | Aug 2017 - Dec 2017

- ▶ Assisted in grant writing, researching, and cultivating grant opportunities for refugee re-settlement programs

TECHNICAL SKILLS

AI & ML	Python/PySpark, R, scikit-learn, XGBoost, random forest, OpenAI API, LLM applications
Neuroimaging	nilearn, CONN toolbox, fMRI pipelines, functional connectivity, CNNs for diagnostic imaging
Cloud Platforms	Azure Databricks, Snowflake, Azure AI Foundry, GitHub, Microsoft SQL Server
Data Visualization	Power BI, Power Platform, Plotly, ggplot2
Domains	Public health & health equity, epidemiology, computational neuroscience, biomedical imaging, EHR data
Methods	Predictive & statistical modeling (ANOVA, regression), prevalence analyses, inter-rater reliability, systematic reviews, hypothesis testing, graph & network analyses

PROFESSIONAL SOCIETIES

2026 - Present	Institute of Electrical and Electronics Engineers (IEEE) ML & Signal Processing, Member
2025 - Present	Organization for Human Brain Mapping, Member
2025 - Present	Alzheimer's Association International Society to Advance Alzheimer's Research and Treatment (ISTAART), Member & Presenter
2023 - 2024	Society for Neuroscience, Member & Presenter
2021 - Present	Black In Neuro, Member & Organizer (2021-2026)